

THE ZERO PARADOX

ZP-E: Bridge Document

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This document is the cross-framework synthesis layer of the Zero Paradox. It imports from ZP-A (lattice algebra), ZP-B (p-adic topology), ZP-C (information theory), and ZP-D (Hilbert space state layer). It provides three formal inserts: DA-1 (Instantiation as Execution), DA-2 (Instantiation Succession), and DA-3 (Perspective-Relative Cardinality). AX-1 (Binary Snap Causality) is retired. Its content was split in two: the shape of the snap is proved, as Theorem T-SNAP, and that the snap occurs is stated separately: it follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2). `tsnap_holds_but_nothing_moves` shows T-SNAP does not carry it. With DA-2, the directed instantiation tree is formally licensed. With DA-3, cardinality is shown to be position-dependent within the instantiation structure.

Illustrated Companion: A paired ZP-E Illustrated Companion document provides accessible explanations and visual summaries of the bridge derivations in this document. Readers new to the framework are encouraged to start with the companion.

A note on framework scope. The four layers (ZP-A through ZP-D) are not claimed to be the only mathematical domains in which the Zero Paradox structure appears. They are chosen because they cover the canonical mathematical languages for describing state: algebraic order structure (ZP-A), topology and metric geometry (ZP-B), information and complexity theory (ZP-C), and functional/Hilbert space analysis (ZP-D). The claim is that the same structural limit appears across each of the four frameworks addressed here — not that these four are the only possible witnesses.

Across all four layers, the same structural limit appears at $\perp$ — but each framework names it differently: ZP-A identifies $\perp$ as the global minimum, below which the lattice's ordering relation cannot descend; ZP-B finds the p-adic valuation undefined (or $+\infty$) at 0, where clopen separation is total — 0 and every nonzero element lie in disjoint clopen classes; ZP-C finds surprisal unbounded above at $\perp$, so no finite external description can contain $\perp$; ZP-D maps 0 to a basis vector orthogonal to every nonzero state, from which no return is possible. DA-1 establishes these as descriptions of the same structural limit across four independent mathematical languages.

Formal Insert DA-1: Derived Proposition — Instantiation as Execution

DA-1 is a Derived Proposition: instantiation as execution, the first Lean formalization of DA-1 (axiom-free, conditional on DP-2, `Snap.lean §VI`), with incompressibility read as self-description (ZP-C D1 + AIT) and CC-2/R3 grounding.

I. The Gap DA-1 Closes

The T-BUF chain from ZP-C established three results:

- L-RUN: The transition $c_0 \rightarrow c_1$ is a nonzero state transition. (ZP-C — Derived)
- TQ-IH: No program outputs $\perp$ without a nonzero intermediate configuration state. (ZP-C — Derived by L-RUN)
- T-BUF: At P_0 , execution is structurally guaranteed; that execution state is ϵ_0 in the semilattice. (ZP-C — Candidate Theorem pending DA-1)

T-BUF was labelled Candidate because Step 2 asserts that a configuration at P_0 is a live machine state — that instantiation at P_0 constitutes an execution event, not a static description. ZP-C L-INF supplies one mathematical premise: $\perp$ at P_0 has unbounded surprisal — no finite external interpreter can hold it as a static description. ZP-A CC-2 supplies a second, structural basis: $\perp = \{\perp\}$ is a self-containing object with no external interpreter by structure. DA-1 (§ IV below) provides the derived proposition that closes T-BUF Step 2.

II. The Two Senses of a Configuration at P_0

Sense A — Descriptive: x exists as a string — a finite syntactic object that has been written down or specified. The machine it describes has not necessarily been instantiated. P_0 is a property of the string. The string is inert.

Sense B — Instantiated: x exists as the current configuration of a running machine. The machine is executing. P_0 is a property of the live configuration. The configuration is active.

III. Design Principle DP-2 — Execution Distinguishability

The two senses of § II share the same output value ($\perp$) but differ in machine state. This separation — output value is not machine state — is made precise by Design Principle DP-2.

Design Principle DP-2 — Execution Distinguishability

Machine states carry execution history independently of output values. A machine in state c_1 can output $\perp$ (the null value) while being in a configuration entirely distinct from a machine in state c_0 . The post-execution null and the pre-execution null are different instances — same output value, different machine state.

Lean formalization (Snap.lean §VI): TrackedOutput separates value : MachinePhase from state : MachinePhase. $\text{preInstantiation} = \langle c_0, c_0 \rangle$; $\text{postInstantiation} = \langle c_0, c_1 \rangle$. Theorem `dp2_execution_distinguishability` proves: $\text{preInstantiation.value} = \text{postInstantiation.value}$ (same output) $\wedge$ $\text{preInstantiation.state} \neq \text{postInstantiation.state}$ (distinct machine states). Proved axiom-free.

Given DP-2, DA-1 is Lean-formalizable at the minimal-path level. Theorem `da1_minimal_path` (Snap.lean §VI) establishes four conjuncts: (1) before and after instantiation, the output value is the same ($\perp$); (2) the machine states are distinct ($c_0 \neq c_1$); (3) the machine was at c_0 before; (4) the machine is at c_1 after. Proved axiom-free — no Kolmogorov complexity, no ZF+AFA required. The "return to null" after instantiation is postInstantiation (output $\perp$, state c_1) — not preInstantiation (output $\perp$, state c_0). Irreversibility (ZPB C3, ZPE `t_snap_irreversible`) blocks any path back to preInstantiation .

IV. Proposition DA-1

Proposition DA-1 — Instantiation as Execution

Claim: The instantiation of a machine configuration c_1 at the incompressibility threshold P_0 is an execution event in the sense of L-RUN. It is not a static description of a machine. It is a machine in state c_1 .

Formal core (Lean): DP-2 (§III) — TrackedOutput separates output value from machine state. `da1_minimal_path` proves: one act of instantiation moves c_0 to c_1 regardless of output value. The returned $\perp$ occupies the bottom role again; reading it as a NEW null rather than the prior c_0 is C-DA2, a modelling commitment. The axiom-free result is the role, not the novelty.

Structural grounding: ZP-A CC-2 ($\perp = \{\perp\}$) establishes $\perp$ as a Quine atom under ZF + AFA — a set that is its own singleton, admitting no external interpreter. ZP-A R3 derives the immediate consequence: a self-containing object cannot be a static description awaiting external instantiation. ZP-C L-INF provides independent informational corroboration: $\perp$ has unbounded surprisal, exceeding the capacity of any finite interpreter.

The three paths below are not parallel proofs of DA-1. Each argues independently for why the precondition DP-2 formalizes holds — why instantiation of $\perp$ at P_0 necessarily constitutes a first instruction fetch rather than a static description. DP-2 + `da1_minimal_path` derive $c_0 \rightarrow c_1$ axiom-free once that precondition is established. The paths are convergent corroboration of the precondition; the formal derivation is DP-2's.

Path 1 — Structural (Self-Execution, ZP-J T-EXEC): Nothing exists outside the null space. $\perp$ is prior to all differentiation — there is no external state, no prior cause, no position from which something else could execute $\perp$. If $\perp$ executes at all, the only possible executor is $\perp$ itself. A thing that executes itself contains itself, and the framework reads this as its own sole member: $\perp = \{\perp\}$, so $\perp \in \perp$. The argument is that this is forced by the impossibility of external execution, which makes it more than a free choice and less than a theorem. ZP-A R3 states the structural consequence: a self-containing object admits no external interpreter position. ZP-J T-EXEC proves that whatever fills the Quine-atom role is $\perp$: $\text{IsQuineAtom}(q) \leftrightarrow q = \perp$, proved axiom-free in Lean 4 (`ZeroParadox.t_exec_iff`). AFA (ZF + AFA) is a consistent set-theoretic home for this structure, and the commitment is not the adoption of AFA specifically: it is a set of requirements on the host theory — that the bottom is a self-containing Quine atom ($\perp = \{\perp\}$), and that this atom is unique — of which AFA is the canonical example. ZP-A CC-2 ($\perp = \{\perp\}$) is a Forced Metatheoretic Commitment: on a lattice, T-EXEC proves that whatever fills the Quine-atom role is $\perp$ (axiom-free, given the field `bot_self_mem`); in AFA set theory $Q = \{Q\}$ is a theorem (Aczel 1988, Example 1.3), and that the framework's $\perp$ is that set is the argued metatheoretic step (see Remark R-AFA).

Path 2 — Informational (ZP-C L-INF): Independently, the surprisal $I(n) = n$ at ball-hierarchy depth n is unbounded — for any finite M , $\exists$ depth n with $I(n) > M$. $\perp$ corresponds to the limit point $0 \in Q_2$; its informational content exceeds every finite bound. Any finite external interpreter can hold only a finite informational bound; $\perp$ exceeds every such bound. Note: Path 2 is motivational context, not a formal path to the conclusion. The step "exceeds every finite bound $\rightarrow$ therefore necessarily executing" is a foundational commitment, not a derivable claim. It asks what it means for a mathematical structure to instantiate rather than merely satisfy conditions — a question no computability library answers. Forward paths: (a) a new axiom explicitly committing to this bridge; (b) a connection to Chalmers' notion of implementation; (c) the dissolution argument in The Philosophical Question That Started This — the separability of description and instantiation is the assumption the framework dissolves, not a gap it must close from the outside.

Path 3 — Formal bridge: Incompressibility as Self-Description (ZP-C D1 + standard AIT): The preceding paths establish that $\perp$ admits no external interpreter. This path provides the formal bridge from that

negative claim to the positive claim (necessarily executing). In the standard Turing model (D7), a machine configuration x exists in one of two states: (A) Static description — x exists as a string specified but not yet being executed; some external program p ($|p| < |x|$) generates x when run, so x is a description awaiting a separate execution event by an external generator. (B) Live execution — x is the current configuration of a running machine. These are exhaustive in the Turing model: either x has a shorter external generator, or it does not. At P_0 , ZP-C D1 gives $K(c_1 |n|)/|c_1| = 1$: c_1 is algorithmically incompressible. No external program p exists with $|p| < |c_1|$ such that $U(p, n) = c_1$. State (A) requires such a p — and no such p exists at P_0 . State (A) is therefore eliminated by the Kolmogorov condition. Since (A) and (B) are exhaustive and (A) is eliminated, c_1 is in state (B): it is executing. Instantiation at P_0 is not the placement of a description to be executed later — there is no shorter prior description to execute. Given DP-2, instantiation and execution are the same act (DA-1).

The formal grounding (DP-2, §III) and the three paths above operate at distinct levels, not as alternatives to one another. DP-2 + `da1_minimal_path` establish a conditional at the level of machine-state representation: if instantiation of $\perp$ constitutes a first instruction fetch in the sense of D7, then $c_0 \rightarrow c_1$ follows necessarily — axiom-free, proved by construction. DP-2 is grounded in D7 itself (the standard computational distinction between before-first-instruction and after-first-instruction states), which is prior to and independent of DA-1. The three paths above operate one level down: they argue for why the precondition holds — why instantiation of $\perp$ necessarily constitutes a first instruction fetch at all. Path 1 (structural) argues that nothing external to $\perp$ can execute $\perp$ — therefore $\perp$ must execute itself, which argues for $\perp = \{\perp\}$ as more than a free choice and less than a theorem (ZP-A CC-2, a Forced Metatheoretic Commitment); ZP-J T-EXEC proves, axiom-free, that whatever fills the Quine-atom role is $\perp$. Path 2 (informational) provides motivational context — unbounded surprisal as a pointer toward why static holding is incoherent — but the bridge from unbounded surprisal (L-INF) to execution is a foundational commitment, not a derived claim. Path 3 (AIT) argues that incompressibility eliminates the static-description alternative. Path 1 is witnessed by `da1_closed_concrete` (ZP-K), which proves `IsQuineAtom( $\perp$  : MachinePhase)` and nothing computational; Path 3's witness is the `machinePhaseKleene` instance's `botCode_is_quine` field, a `KleeneStructure` requirement, not a second independent proof. The two are carried together by `da1_paths_unified` as a conjunction of witnesses; that they name one structural fact is the framework's reading. Path 2 identifies a missing principle; its forward resolution is in *The Philosophical Question That Started This*. Path 2 is context. (ZP-M R-M.1 provides a retrospective structural analysis of why the gap resisted formalization.)

Status: DERIVED PROPOSITION — primary formal grounding: DP-2 (§III, `TrackedOutput` construction). `da1_minimal_path` proved axiom-free in Lean (Snap.lean §VI): instantiation moves c_0 to c_1 regardless of output value. ✓ Path 1 (structural, ZP-J T-EXEC + ZP-K): witnessed by `da1_closed_concrete` : `IsQuineAtom( $\perp$  : MachinePhase)`, proved in `Kleene.lean`, which proves nothing computational. (Under `MachinePhase`'s `selfMem x := x =  $\perp$` , this reduces to $(\perp = \perp) \wedge (\forall x, x = \perp \Rightarrow x = \perp)$ — structural closure enforced by typeclass design, not a set-theoretic derivation from ZF+AFA. See R-K.0.) Path 2 (informational, L-INF): FOUNDATIONAL COMMITMENT — a missing principle, not a missing proof. Forward: *The Philosophical Question That Started This*; ZP-M R-M.1 (retrospective structural analysis). Path 3 (computational, ZP-K Kleene): its witness is the `machinePhaseKleene` instance's `botCode_is_quine` field — a `KleeneStructure` requirement, not a second independent proof; `da1_paths_unified` carries it with Path 1 as a conjunction of witnesses. CC-1 ($S_0 = \perp$): restated — ZP-J `cc1_derived` with `t_exec_iff` makes a Quine-atom start and a $\perp$ start the same condition (axiom-free, Lean) — and not forced: on a carrier with a second point a valid sequence starts elsewhere (`ZeroParadox/Settheory/OntBridge.lean`). CC-2 ($\perp = \{\perp\}$): ZP-J `t_exec_iff` proves $\perp$ is the only occupant of the Quine-atom role (axiom-free); that $\perp$ is the AFA set $Q =$

{Q} is an argued metatheoretic commitment (see R-AFA). DP-2 (§III) — explicit. T-SNAP’s premises, in its Lean form and in its prose argument, are stated under Premises of T-SNAP (DA-1 insert, § V). AIT (Kolmogorov complexity) outside Lean scope; the Kleene structure is its in-scope counterpart, carried as a KleeneStructure requirement (botCode_is_quine), not a proof of the AIT claim.

V. Theorem T-SNAP — Binary Snap Causality

Theorem T-SNAP — Binary Snap Causality

“Causality” in this name refers to the shape of the step, not to its occurrence.

Statement: The Binary Snap $\perp \rightarrow \varepsilon_0$ is a derived consequence of P_0 , L-RUN, TQ-IH, DA-1, and ZP-A D2, among the results the seven steps below cite. It is not an axiom. What it rests on differs between its Lean form and its prose argument: see Premises of T-SNAP, directly below.

Premises of T-SNAP, in two readings. Both readings are true, and neither replaces the other. In Lean, two of the premises, CC-1 and occurrence at the first step, are the hypotheses of `t_snap_given`, named in (i).

- (i) Lean form: the shape. `t_snap_derived` : $c_0 \neq c_1 \wedge c_1 \neq c_0 \wedge \text{join } c_0 \ c_1 = c_1$ (ZeroParadox/Order/Snap.lean) takes no hypotheses, and `#print axioms` reports no axioms. It is a fact about the fixed two-state type `MachinePhase`, with the initial phase c_0 as its bottom element $\perp$ and the running phase c_1 above it: the two phases differ, and joining them gives c_1 . Nothing is passed to it as an argument; binary existence is built into the choice of that type, and the start at $\perp$ into the statement’s choice of c_0 . It fixes the SHAPE of the transition, not that one occurs: `tsnap_holds_but_nothing_moves` proves the same statement in a dynamics where every phase is a fixed point. Two of the premises as hypotheses: `t_snap_given` (same file) holds over any join-semilattice with bottom, taking $S_0 = \perp$ (hcc1, CC-1) and $S_1 \neq S_0$ (hocc, occurrence at the first step: the step at index 1 is taken) and giving the same shape at S_0, S_1 . Its two inequalities restate hocc; only the join conjunct is derived, from A4. `t_snap_derived`’s statement is its `MachinePhase` instance at the sequence $c_0, c_1, c_1, \dots$, where hcc1 holds by `rfl` and hocc reduces to $c_1 \neq c_0$. `MachinePhase` does not discharge occurrence: the sequence that stays at c_0 is a state sequence in the same type and fails hocc. With no dynamics given, dropping either hypothesis makes the conclusion fail (examples beside it). hocc forces at least two points and does not force exactly two; AX-B1’s discreteness is not in that signature.
- (ii) Prose argument: the shape, then the occurrence. The seven steps below read that shape as a statement about the framework’s own state sequence. That reading uses commitments, among them AX-B1 (binary existence, ZP-B, the framework’s one substantive modelling commitment) at Step 4, and CC-1 ($S_0 = \perp$, a ZP-A Conditional Claim) for the start at $\perp$ in Steps 4 and 6. That the transition OCCURS rests further on two things. The occurrence commitment: instantiation occurs (a machine configuration reaches P_0). And Step 2, which imports DA-1: a configuration at P_0 is executing. DA-1 is closed only given DP-2: § IV derives $c_0 \rightarrow c_1$ from DP-2 once DP-2’s precondition is established, and calls its three paths convergent corroboration of that precondition; Path 1 argues through ZP-A CC-2, itself a commitment. Together they give that the Snap occurs, which is not a theorem. Given CC-1, its first-step form is the hypothesis hocc of `t_snap_given` (a sequence that first moves at a later index fails hocc). ZP-C L-INF supplies unbounded surprisal, and the step from there to forced execution is a foundational commitment (§ IV), not a mathematical consequence of L-INF alone.

Proof:

Note on Lean scope: the formal Lean proof of T-SNAP is a single term $\langle l_run, tq_ih, rfl \rangle$ — l_run proves $c_0 \neq c_1$ by decide, tq_ih is its symmetric form, and rfl closes the join in the MachinePhase semilattice (t_snap_given obtains that conjunct from A4, bot_join). What follows is the informal motivation for the modelling choices that let that term be read as a statement about the framework's own state sequence.

- Step 1 — P_0 identifies the incompressibility threshold. When $K(x|n)/n = 1$, the configuration string x is algorithmically random. (ZP-C D1)
- Step 2 — A configuration at P_0 is necessarily executing: At P_0 , $K(c_1|n)/|c_1| = 1$ (ZP-C D1) — c_1 is incompressible, its own minimal program. No shorter external generator exists; the static description state is eliminated; c_1 is in live execution (DA-1 Path 3 — from ZP-C D1 + AIT). Corroboration: $\perp = \{\perp\}$ (ZP-A CC-2/R3); unbounded surprisal (ZP-C L-INF). (DA-1 § IV — Derived Proposition)
- Step 3 — Any instantiated execution passes through c_1 . (ZP-C D7 — definitional; c_1 is the first running configuration)
- Step 4 — $c_1 \neq \perp$. (ZP-C L-RUN — Derived; c_1 has gained execution context not present in $c_0 = \perp$; by AX-B1 this is a distinct, nonzero state)
- Step 5 — No program that executes produces only $\perp$ configuration states. (ZP-C TQ-IH — Derived; execution trace $\tau(p)$ contains c_1 for any executing program p)
- Step 6 — In $(L, \vee, \perp)$, c_1 is an element strictly above $\perp$. By ZP-A D2, the transition $\perp \rightarrow c_1$ is a valid state transition: $c_1 = \perp \vee \epsilon_0$ for some $\epsilon_0 \in L$ with $\epsilon_0 > \perp$. This transition is the Binary Snap.
- Step 7 — The transition is irreversible: algebraically by ZP-A R1 (no subtraction operator); topologically by ZP-B C3 (no continuous return path to 0 in Q_2). These two grounds are sufficient. Conceptual correspondence: ZP-G AX-G2 ($\text{hom}(X, 0) = \emptyset$ for $X \neq 0$) expresses the same irreversibility in categorical language — ZP-G is downstream of ZP-E and is not a formal dependency of this proof.

Conclusion: The Binary Snap is a derived consequence, given the commitments under Premises of T-SNAP. AX-1 (Binary Snap Causality) is retired. Its content was split in two: the shape of the snap is proved, as Theorem T-SNAP, and that the snap occurs is stated separately: it follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2). $t_snap_holds_but_nothing_moves$ shows T-SNAP does not carry it. ✓

Status: DERIVED — Cross-Framework. Premises: in two readings, under Premises of T-SNAP above; the Lean form fixes the shape with no hypotheses, t_snap_given takes CC-1 and occurrence at the first step as hypotheses, and the prose argument's shape and its occurrence rest on commitments, among them those named there. Dependencies cited in the steps: ZP-C D1, D7, L-RUN, TQ-IH; ZP-B C3; ZP-A D2, R1; ZP-E DA-1; ZP-J T-EXEC. CC-1 ($S_0 = \perp$): restated — a Quine-atom start and a $\perp$ start are one condition (ZP-J $cc1_derived$, t_exec_iff ; axiom-free) — and not forced, since on a carrier with a second point a valid sequence starts elsewhere. CC-2 ($\perp = \{\perp\}$): ZP-J t_exec_iff proves $\perp$ is the only occupant of the Quine-atom role (axiom-free); that $\perp$ is the AFA set $Q = \{Q\}$ is argued metatheoretically (see R-AFA). Both remain commitments: CC-1 is a Conditional Claim, expressed through the Quine-atom role and not removed by it; CC-2 is a Forced Metatheoretic Commitment. Conceptual correspondence only: ZP-G AX-G2 (downstream of ZP-E; not a formal dependency).

Remark R- ϵ_0 — On the Symbol Choice for the Snap Displacement

Note: this remark draws an informal structural analogy. No formal embedding of ZP's ϵ_0 into the ordinal ϵ_0 is claimed or established here. The symbol ϵ_0 in Step 6 names the state the Binary Snap reaches from $\perp$: by Step 6 and A4 ($\perp \vee \epsilon_0 = \epsilon_0$), $\epsilon_0 = c_1$. Nothing in Step 6 makes ϵ_0 the least state above $\perp$, or even a state with nothing between it and $\perp$; on the two-state carrier MachinePhase it is both, since c_1 is the only state above c_0 . This symbol is chosen deliberately to coincide with ϵ_0 , the proof-theoretic ordinal of Peano Arithmetic.

The ordinal ϵ_0 (Cantor 1897, Math. Ann. 49, §20, Satz B). In ordinal arithmetic, ϵ_0 is the smallest fixed point of the map $\alpha \rightarrow \omega^\alpha$: equivalently, $\epsilon_0 = \sup\{\omega, \omega^\omega, \omega^{\omega^\omega}, \dots\}$. No finite ω -tower reaches it. Gentzen proved in 1936 that transfinite induction up to ϵ_0 is sufficient to prove Con(PA). Necessity — that no smaller ordinal would do — is the other direction, and it rests on different ground: the provability in PA of transfinite induction at every ordinal strictly below ϵ_0 , which Gentzen reports as already known, crediting Hilbert-Bernays, together with Gödel's second incompleteness theorem, by which PA cannot prove Con(PA) from within. The ordinal ϵ_0 is therefore the minimum threshold at which finite arithmetic exhausts its own generative capacity.

The structural analogy. ZP's ϵ_0 occupies an analogous position in the state lattice. At P_0 , c_1 satisfies $K(c_1 | n) / |c_1| = 1$ (ZP-C D1): it is algorithmically incompressible — no finite external program shorter than c_1 generates it. Just as the Cantor ϵ_0 cannot be reached from 0 by any finite ω -tower, ZP's ϵ_0 cannot be reached from $\perp$ by any finite external description. Both name a structurally analogous object: a witness for a transition that exhausts the finite generative hierarchy below it.

The proof structures are analogous. The proof-theoretic analysis locates the minimum ordinal up to which PA cannot prove transfinite induction. ZP locates a state displacement at which no external program can hold $\perp$ as a static string — where unbounded surprisal (ZP-C L-INF) and self-containment (ZP-A CC-2) together eliminate any external interpreter position — and, unlike the ordinal side, nothing makes it the least such displacement (see the first paragraph). In both cases the exhaustion of finite description is not a deficiency but a structural consequence: the system is necessarily executing at ϵ_0 .

What is not claimed. ZP does not assert that L is an ordinal structure, or that ZP's ϵ_0 is literally the Cantor ordinal under a formal embedding into the p-adic/lattice framework. The analogy is motivational: both ϵ_0 s mark a witness for incompressibility relative to a finite base. A formal embedding — showing that the Cantor ϵ_0 is order-isomorphic to or embeds into the p-adic completion of L at $\perp$ — remains an open question and would constitute a strengthening of this claim.

Remark R-AFA — Why Foundation is Ruled Out; Non-Well-Foundedness Forced, AFA Adopted

CC-2 — the identification of $\perp$ with the Quine atom $\perp = \{\perp\}$ — is a set-theoretic step beyond ZP-A's algebra: it is not derived from A1–A4. What it entails is nonetheless not arbitrary, and the two halves separate. The membership structure the framework's bottom must have forces a non-well-founded host, which is what rules ZF + Foundation out. Which non-well-founded axiom to adopt is a further step that the forcing does not take: ZF + AFA is the host this framework adopts, as the canonical theory meeting the requirements, not the one theory the bottom selects.

Remark R-AFA — Why Foundation is Ruled Out; Non-Well-Foundedness Forced, AFA Adopted

Foundation rules out the bottom (structural). The framework's bottom is the Quine atom $\perp = \{\perp\}$: it is a member of itself, $\perp \in \perp$. ZF + Foundation (the Axiom of Regularity) forbids exactly this — under a well-founded membership relation no set lies on a membership cycle of any length, so in particular no set is a member of itself. This is a theorem of ZF + Foundation, not a framework-specific stipulation; it is machine-checked here as `no_quine_atom / no_membership_cycle` (ZeroParadox/Settheory/Wall.lean, footprint [propext, Quot.sound], choice-free). A Foundation-respecting universe therefore cannot host $\perp$ at all.

The framework's readings of $\perp$ support the premise that $\perp$ is circular; the exclusion itself is a theorem. R3 (ZP-A) — $\perp = \{\perp\}$ has no describer position external to itself — and L-INF (ZP-C) — the 2-adic surprisal $I(n) = n$ is unbounded, so no finite interpreter can hold $\perp$ — each describe an object with no external vantage and unbounded self-referential depth. That is exactly the profile of a non-well-founded, self-membered set: the two readings motivate why $\perp$ must be the circular object Foundation excludes, rather than serving as independent proofs of the exclusion.

Non-well-foundedness is forced; AFA is the canonical replacement. The membership structure of $\perp = \{\perp\}$ is circular ($\perp \in \perp \in \perp \in \dots$), so $\perp$ can live only in a non-well-founded metatheory. That much is forced; the particular axiom is not. Aczel records four anti-foundation axioms, and AFA, FAFA and SAFA each admit exactly one Quine atom — his own term is a reflexive set — while being pairwise incompatible (Aczel 1988, Corollary 4.28; Appendix A, p. 108). Supplying exactly one therefore does not single AFA out, so AFA is adopted as the canonical witness rather than selected by the membership structure.

What is machine-checked (the requirements framework). The question of which host theory can admit $\perp$ is reduced to the two requirements a host theory must supply — (Y) a Quine atom $\perp = \{\perp\}$ and (Z) its uniqueness — and largely closed in the framework's set-theory layer (the QuineHost development, ZeroParadox/Settheory/QuineHost.lean). Proved there: the third property, (X) Foundation-freeness, is not assumed — it is forced by (Y) (`quineHost_not_wellFounded`, axiom-free); ZF + Foundation is excluded (`zfSet_no_quine_bottom` — no set is self-membered — not axiom-free: it inherits `no_quine_atom`'s footprint [propext, Quot.sound], choice-free); and the requirements are realizable — a concrete one-atom model exhibits them (`oneAtom_not_wellFounded`, axiom-free). AFA is the canonical example, exhibited off the framework's own `AFAStructure` (`afaStructure_isQuineHost`) — a definition rather than a theorem about the set theory, and one every ZP-A lattice supplies trivially (the NO-GO gauge in ZeroParadox/Settheory/SetTheoryAFA.lean). So the reduction to requirements is not merely argued: the forcing and the realizability are settled results. What stays argued is which requirements to demand, below.

Remark R-AFA — Why Foundation is Ruled Out; Non-Well-Foundedness Forced, AFA Adopted

Boffa is set aside on a weaker warrant, and the cited half and the checked half are separate. That Boffa's anti-foundation axiom admits a proper class of Quine atoms rather than one is a cited fact about that theory, and it reaches that axiom through a weaker one, in a direction worth stating. Boffa's axiom is equivalent to his earlier axiom BA_1 together with an extension condition, so it implies BA_1 (Aczel 1988, Exercise 5.14(i) for the equivalence; p. 59, where the axiom is formulated as Boffa's strengthening of BA_1); and BA_1 on its own already forces the reflexive sets — Aczel's term for Quine atoms — to form a proper class (Aczel 1988, Proposition 5.1). That permissiveness is therefore inherited from the weaker axiom rather than introduced by the strengthening. This development formalizes no part of either Boffa axiom. What is machine-checked is the mechanism behind it — that self-membership alone does not force uniqueness: in a two-element toy model where every element is self-membered, no element is the self-membered one (`boffa_fails_unique`, `ZeroParadox/Settheory/QuineHost.lean`, axiom-free). That is a counter-model separating the two requirements, and it blocks one direction only — a host supplying the Quine atom need not supply its uniqueness. Dropping uniqueness accordingly leaves a host theory no less consistent; what it leaves standing is the Quine-atom clause alone, and in that toy model every element satisfies that clause, so the surviving clause is uninformative as a pinning condition — every element is a witness, and none is pinned as the host's distinguished bottom. Neither half is an in-kernel result about Boffa's axiom itself.

What remains — a narrow Forced Metatheoretic Commitment. With the forcing and realizability proved, one step is not a theorem: that a Quine atom and its uniqueness are the right two requirements to demand — the commitment to $\perp = \{\perp\}$ as the minimal option consistent with A4's additive-identity role (it has exactly one member, itself, and introduces no internal differentiation; any extension $\perp = \{\perp, x\}$ with $x \neq \perp$ would add members carrying their own membership chains). This is a Forced Metatheoretic Commitment — stronger than a free choice, weaker than a theorem — not a Conditional Claim; the set-membership face $\perp \in \perp$ stays metatheoretic, while the occupant of the Quine-atom role is machine-checked to be $\perp$ (`t_exec`, `ZeroParadox/Settheory/SetTheoryAFA.lean`, axiom-free, given the field `bot_self_mem`). The named falsifier is correspondingly narrow: a differently-motivated requirement set that hosts $\perp$ without demanding a unique Quine atom would reopen this choice — it would not touch the forcing or the realizability, which are settled.

VI. Effect of T-SNAP on Downstream Results

Remark R-DA1: AX-1 is retired, and its content was split in two. Results in ZP-E that previously depended on AX-1 as an axiom now depend on T-SNAP, a derived theorem, for the shape, and, wherever they use that the Snap happens, on the occurrence commitment (instantiation occurs) together with DA-1. T5 (Iterative Forcing Theorem) depended on AX-1 and is restated below. T4 (Unified Snap Description) carried AX-1 as an axiom label on the causality component — that label is upgraded to Derived — T-SNAP for the shape, with occurrence still a commitment. DA-1 is additionally upgraded from Design Principle to Derived Proposition, closed given DP-2 (`da1_minimal_path`); ZP-A CC-2 ($\perp = \{\perp\}$) and R3 motivate it, with ZP-C L-INF as independent corroboration. The intentional axioms of the system are now: AX-B1 (binary existence), AX-G1 (initial object), AX-G2 (source asymmetry). AX-1 is retired: its shape is Theorem T-SNAP, and the occurrence commitment is not on this list because it is a commitment, not a named axiom. Additional structural commitments are carried as typeclass fields: A4 (`bot_join`, `ZPSemilattice` — standard semilattice algebra), `AFAstructure.quine_unique` and `bot_self_mem` (the Quine-atom role), `KleeneStructure.botCode_is_quine` (computational closure). Like named axioms these are assumptions, carried as hypotheses by every theorem that uses them; `#print axioms` does not surface typeclass fields. Unlike named axioms they can be cheap to meet: every ZP-A lattice supplies `AFAstructure` trivially.

T5 (Iterative Forcing Theorem) — restated

T5 (restated). Selection. In the ordinal chart the next rung is the snap re-seeded one step past the current one (succession_succ), and none lies between two consecutive rungs (ZeroParadox/Ordinal/SnapSuccession.lean § I). Carried into the two-state lattice by a monotone map that sends the tower's stages to c_0 and ε_0 to c_1 , no ordinal below ε_0 is sent to c_1 , and every fixed point of $\alpha \mapsto \omega^\alpha$ is (snap_unconditional, hfp_from_epsilon_zero). The two faces of ε_0 carry the two directions: as the supremum of the stages, every ordinal below ε_0 lies under one of them, so nothing below fires (fundamentalSeq_cofinal); as the least fixed point of $\alpha \mapsto \omega^\alpha$ it lies below every other, so every fixed point of $\alpha \mapsto \omega^\alpha$ fires (hfp_from_epsilon_zero, epsilon0_min_eq_max). The placement at ε_0 is a hypothesis there ($h\varepsilon_0$), which the Lean names the alignment hypothesis; deriving it from the 2-adic structure is open, as the Classical.choice inversion conjecture (ZeroParadox/Ordinal/Incompleteness.lean). The rungs are the iterative bottoms; none of them is $\perp$ (epsilon0_ne_bot; for every rung, Ordinal.epsilon_pos). On an arbitrary lattice no step is selected: 0, 2, 4, ... on the natural numbers satisfies A1–A4 and AX-B1.

Iteration. No step is guaranteed. A state sequence moves only upward (T3) and never returns (R1, t_snap_irreversible). Where the next increment is already absorbed ($S_n \vee \alpha_n = S_n$), the state stays the same, which T-SNAP permits (tsnap_holds_but_nothing_moves); different runs need not reach the same height. If from some step x on every new increment is already absorbed, then S_x is the supremum of the run: the value that run reaches in the limit. Short of a top, reaching it cannot be confirmed at any finite step; a run that reaches a top state (c_1 on the two-state carrier) is known there to stay.

Reading: ε_0 is both at once for the ordinal tower: the least fixed point of $\alpha \mapsto \omega^\alpha$ and the supremum of its stages (epsilon0_min_eq_max).

The handle T5 is kept. “Forcing” names the shape of each step taken, not that any step is taken.

Formal Insert DA-2: Instantiation Succession — The Multiple- $\perp$ Result

Formally licenses the directed instantiation tree

I. The Gap DA-2 Closes

Given the premises stated under Premises of T-SNAP (DA-1 insert, § V), DA-1 and T-SNAP give the Binary Snap on reaching P_0 within an instantiation. Three questions remain open after DA-1:

- CC-1 (ZP-A) posits $S_0 = \perp$ — not derivable from A1–A4 alone (ZP-J restates it as an equivalence with starting at a Quine atom, cc1_derived and t_exec_iff, and does not force it). This leaves open whether $\perp$ is unique across all instantiations or whether each instantiation carries its own $\perp$.
- ZP-B R1 distinguishes universal structure from universe-contingent parameters. ε_0 is contingent per instantiation. Whether $\perp$ is similarly contingent is not addressed in ZP-A through ZP-D.
- Occurrence fence. T-SNAP fixes the SHAPE of each step. It does not establish that any step is taken: tsnap_holds_but_nothing_moves exhibits a model in which T-SNAP holds and nothing moves. Every “fires” below narrates the commitment that instantiation occurs, not a consequence of the theorem.
- T-SNAP fires wherever P_0 conditions are met. If the terminal state of instantiation I_n satisfies P_0 conditions, T-SNAP should apply — but this requires formally connecting that terminal state to a $\perp$ role-occupant, read as a new $\perp$ under C-DA2. DA-2 provides this connection.

II. Why ZP-B C3 is Not Violated

C3 prohibits a continuous path from $x \neq 0$ back to the same 0 in Q_2 . DA-2 does not require a return path. The irreversibility of C3 is preserved within each instantiation. What crosses the instantiation boundary is not a path in Q_2 — it is the generation of a new Q_2 with its own metric, its own $\perp$, its own ε_0 . C3 quantifies only over paths within a single topological space and has nothing to say about the boundary between spaces.

More precisely: C3 rules out a continuous return to 0 WITHIN one Q_2 . It does not follow that a recurrence of $\perp$ must be a structurally distinct $\perp$ — that step needs the separateness of the spaces, which is a modelling commitment (C-DA2) and not something the topology supplies. The topology does not enforce novelty; it is silent on it, and in the 2-adic chart the arc reapproaches the same 0 (snap_arc_z2_loop).

III. Definitional Alignment DA-2 — Instantiation Succession

Definitional Alignment DA-2 — Instantiation Succession

Claim: A state S in instantiation I_n satisfies the structural role of $\perp$ for instantiation I_{n+1} if and only if it satisfies A4 relative to all subsequent joins in I_{n+1} :

$S \vee x = x$ for all x in the semilattice of I_{n+1} .

Grounding: A4 is the load-bearing axiom of ZP-A — it defines $\perp$ as the additive identity under $\vee$, the element that contributes nothing to any join and is therefore present in everything above it. DA-2 does not redefine $\perp$. It clarifies that the CC-1 identity condition ($S_0 = \perp$) can be satisfied by any state meeting A4's algebraic conditions — not only by a cosmologically primitive $\perp$. The identity condition is structural, not historical: what matters is the algebraic role a state plays in the subsequent semilattice, not where it came from.

The terminal state of I_n arrives at I_{n+1} carrying the accumulated join of everything in I_n 's sequence. It is structurally $\perp$ to I_{n+1} — contributing nothing to subsequent joins — while being informationally rich relative to I_n . This is the Zero Paradox instantiated at the inter-instantiation level: the terminal state is simultaneously a terminus and a foundation.

Status: DEFINITIONAL ALIGNMENT — no new axiom introduced. DA-2 is a clarification of the scope of CC-1 and A4. ✓

IV. Conditional Claim C-DA2 — Novelty of Successive $\perp$

Conditional Claim C-DA2 — Novelty of Successive $\perp$ (a modelling commitment)

Commitment: each instantiation carries its OWN topological space, so the $\perp$ of I_{n+1} is not an element of the Q_2 of I_n . GIVEN that, no two instantiations share a $\perp$, and succession is a chain (or tree) rather than a cycle.

Status: CONDITIONAL, not derived. The commitment above is the separateness of the spaces; it is assumed, not obtained from C3. Formerly labelled a Corollary and marked derived — retracted, see the note below. The handle C-DA2 is unchanged.

What IS proved: that anything occupying the bottom role IS the bottom of its own lattice (t_iz_limit_is_new_null). That is an identity with the bottom already present, not the production of a second one.

Why this is NOT a proof, stated plainly because it was published as one. The step "the $\perp$ of I_{n+1} is an element of a distinct topological space" is the CONCLUSION, not a consequence of C3. C3 quantifies over paths WITHIN a single space — it says no continuous path runs from any $x \neq 0$ back to 0 inside one Q_2 — and has nothing to say about whether the next instantiation's space is a different one. Assuming separateness and then deriving distinctness from it is circular. DA-2 likewise fixes identity conditions WITHIN an instantiation; it does not compare across two.

And the corpus measures against it where the question is concrete: in the 2-adic realization the arc returns to the SAME zero. `snap_arc_z2_loop` proves the tower starts at 0, that every finite stage is $\neq 0$, and that it converges back to 0; `tower_image_loops_to_seed` states the limit IS the seed value. So in that chart novelty does not merely lack a proof — it fails. C-DA2 stands as a modelling commitment about instantiations, and must not be cited as derived, nor supported by T-IZ or by `t_iz_limit_is_new_null`.

V. The Directed Instantiation Tree

With DA-2 and C-DA2 in place, the global structure of instantiations is a forward-directed tree with no back edges:

- Each node in the tree is a $\perp$ — the bottom element of one instantiation and the foundation of all successor instantiations branching from it.
- Each edge within an instantiation is a step in a monotone state sequence (ZP-A T3). Edges are irreversible (ZP-B C3).
- Branching at each node: every distinct outbound vector from the terminal state of I_n is a valid ϵ_0 for a distinct I_{n+1} . T-SNAP does not select among branches. Because $\perp = \{\perp\}$ (ZP-A CC-2) is the single self-containing $\perp$ with no internal differentiation, every ϵ_0 that represents a first differentiation in any direction is a valid outcome. The work here is done by the undifferentiated $\perp$ of CC-2, not by T-SNAP: given that instantiation occurs, no direction is privileged, so the structure branches rather than threading a line. T-SNAP fixes the shape of each step; it does not supply the occurrence.
- No back edges: C-DA2 COMMITS to no instantiation reaching the $\perp$ of any ancestor. This is the commitment, not a theorem - in the 2-adic chart the arc returns to the same 0 (`snap_arc_z2_loop`).

Remark R-DA2: T-SNAP fires wherever P_0 conditions are met. DA-2 establishes that the terminal state of I_n satisfies those conditions for I_{n+1} . T-SNAP therefore applies across instantiation boundaries without modification. No new axiom is required. The multiverse is not the claim that T-SNAP fires in all directions simultaneously: it is the claim that $\perp = \{\perp\}$ (ZP-A CC-2) has no internal differentiation and therefore no preferred direction for ϵ_0 . The multiverse of instantiations is the full set of all minimal differentiations available from the single self-containing $\perp$ — a structural consequence of CC-2 + T-SNAP + DA-2 jointly.

VI. The Zero Paradox Iterated

The paradox of $\perp$ — simultaneously contributing nothing and being present in everything — propagates structurally at every branching node of the tree. Each node is:

- Nothing to its successor instantiations: it acts as additive identity under $\vee$, contributing nothing to any subsequent join.
- Everything to its successor instantiations: every state in I_{n+1} satisfies $\perp_{n+1} \leq S$, so the node underlies everything that follows.

The tree is the geometric shape of the Zero Paradox iterated across instantiations. The single-instantiation linear sequence was always a cross-section of a structure with this shape.

The complete picture: The Zero Paradox describes a forward-directed infinite tree where $\perp_1 \rightarrow \dots \rightarrow S_{\text{terminal}}^1 \equiv \perp_2 \rightarrow \dots$, where $\equiv$ means structurally satisfies the role of, not is identical to. Each arrow within an instantiation is monotone and irreversible. Each $\equiv$ crossing is not a path — it is a new instantiation of the universal structure.

VII. Implications Within the Framework

Structural Implication: Branching Tree Structure. T-SNAP fixes the SHAPE of the Binary Snap — that if $\perp$ transitions it goes to some $\varepsilon_0 > \perp$, and that the step is one-way. It does not establish that the transition is taken: that the Snap occurs follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2), and the NO-GO gauge `tsnap_holds_but_nothing_moves` exhibits a model in which T-SNAP holds and nothing moves. DA-2 establishes that any terminal state satisfying P_0 conditions acts as $\perp$ for a successor instantiation, generating a forward-directed branching tree. The branching comes from CC-2's undifferentiated $\perp$ and the succession from DA-2, both conditional on instantiation occurring. Note: T-SNAP alone does not establish that it fires on all outbound vectors simultaneously — that universality is the scope of DA-2, not a direct consequence of the snap theorem itself.

Monotonicity and Path Irrecoverability. Within an instantiation, state sequences are monotone — no state can be decreased (ZP-A R1). Every state change is a join operation: $S_n \vee \alpha$ for some increment α . The algebra constrains only that the sequence be monotone, not which monotone path is taken. Each join adds informational content irreversibly. States are permanently encoded in the element's position in L.

Ordinal Direction of State Sequences. The monotone sequence (ZP-A T3) is a structural definition of directional asymmetry in state ordering. Irreversibility is C3 applied within an instantiation. The framework does not assume directional asymmetry — it derives it.

Causal structure. Every state is fully determined by the joins that produced it. The causal history of any state is encoded in its position in L. No effect without the join that produced it.

Formal Insert DA-3: Perspective-Relative Cardinality

Cardinality as position-dependent measurement within the instantiation structure

I. The Gap DA-3 Closes

DA-2 establishes that instantiations form a directed tree and that branching at each $\perp$ node produces multiple successor instantiations. This raises a question about the cardinality of branching: is the fan at each node countably or uncountably infinite? The answer, which DA-3 formalises, is that this question is perspective-dependent — and DA-3 further conjectures (Claim DA-3-C1, deferred to OQ-E2) that this perspective-dependence is the same structural feature underlying the major cardinality anomalies of classical set theory.

II. Perspective-Dependence of Branching Cardinality

From within instantiation I_n , an observer occupies exactly one branch of I_{n+1} . The other branches of I_{n+1} are not accessible via any path — C3 and monotonicity jointly prohibit it. From inside, the branching factor is always 1: the observer sees one branch. From outside — from a meta-level view of the tree — the

branching factor is the full fan of accessible outbound vectors.

Definition DA-3-D1 — Accessible Cardinality

The accessible cardinality of a position p in semilattice L is the cardinality of the set of states reachable from p by monotone sequences within the instantiation containing p .

The accessible cardinality from p is determined entirely by the structure of L above p . It is not an intrinsic property of a collection — it is a property of the relationship between a position and the states reachable from it. No position within any instantiation can access all cardinalities simultaneously. The meta-level view, which sees the full branching fan, is not a position any element of any instantiation can occupy.

Remark R-DA3-1: To observe the full branching fan, one would need to occupy a position outside all instantiations. That position would itself be a state in some semilattice, subject to the same rules. The meta-view is either another instantiation (in which case the tree has no privileged outside view) or $\perp$ itself (in which case $\perp$ is the only position from which the full structure is visible — the state that contributes nothing and is present in everything). The Zero Paradox’s name is more precise than it first appeared.

III. The Cardinality Hierarchy as Perspective-Relative

Cantor’s theorem establishes that for any set S , $|P(S)| > |S|$, generating the hierarchy $\aleph_0 < 2^{\aleph_0} < 2^{2^{\aleph_0}} < \dots$ DA-3 reframes this hierarchy not as a fixed ladder that mathematics climbs, but as a perspective-relative description of the branching structure of the instantiation tree, as seen from within different positions.

Claim DA-3-C1 — Perspective-Relative Absolute Cardinality

The appearance of absolute cardinality — cardinality as an intrinsic property independent of measuring position — is an artifact of treating the semilattice as having a view from outside. DA-2 and C-DA2 jointly prohibit such a view from within any instantiation.

The candidate claim (DA-3-C1) is that accessible cardinality from within any instantiation cannot replicate the view from outside. Whether specific independence results in classical set theory are formal expressions of this perspective-dependence is the conjecture that OQ-E2 is tasked with investigating.

Status: DEFINITIONAL ALIGNMENT + CANDIDATE CLAIM. DA-3-D1 and R-DA3-1 are definitional. DA-3-C1 is a candidate claim: structurally motivated within the framework; formal derivation of the connection between accessible cardinality and specific set-theoretic independence results is deferred to OQ-E2.

Updated Open Items Register

Item	Status	Description
AX-1: Binary Snap Causality	RETIRED — shape proved as T-SNAP; the snap occurs given the occurrence commitment and DA-1	Its content was split in two: the shape of the Snap is proved, as Theorem T-SNAP (from L-RUN, TQ-IH and the bottom law, with no Lean kernel axioms), and that the Snap occurs is stated separately: it follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2). Premises under Premises of T-SNAP (DA-1 insert, § V); given CC-1, the first-step form of the Snap occurring is the hypothesis hocc of t_snap_given.

Item	Status	Description
DA-1: Derived Proposition (DP-2 formal grounding)	CLOSED — DP-2 (formal core); CC-2 + L-INF + AIT (corroboration of precondition)	Primary formal grounding: DP-2 (TrackedOutput, Snap.lean §VI) — <code>da1_minimal_path</code> proved axiom-free. Instantiation of $\perp$ moves machine from c_0 to c_1 ; output value is irrelevant to whether execution occurred. Informal corroboration: ZP-A CC-2 + R3 (structural); ZP-C L-INF (informational); AIT incompressibility (Path 3).
DA-2: Instantiation Succession	CLOSED — Definitional	Terminal state of I_n satisfies A4 role of $\perp$ for I_{n+1} . C-DA2 COMMITS to the novelty of each $\perp$; it does not establish it.
DA-3: Perspective-Relative Cardinality	CLOSED (definitional) / CANDIDATE (DA-3-C1)	DA-3-D1 establishes accessible cardinality is position-dependent within the instantiation structure. DA-3-C1 (candidate): no position within an instantiation can replicate the outside view. Whether this connects formally to specific set-theoretic independence results is deferred to OQ-E2.
OQ-A1: Increment selection	OQ-A1b CLOSED by A1–A4; OQ-A1a: no reason to restrict to join-irreducibles (well-founded carriers); placement in the ordinal tower, for a monotone map sending the tower's stages to c_0 , conditional on the alignment hypothesis $h\varepsilon_0(\varepsilon_0$ sent to $c_1)$	T5 (Iterative Forcing Theorem), restated in DA-1 insert § VI. No step is guaranteed. On an arbitrary lattice no step is selected: 0, 2, 4, ... on the natural numbers satisfies A1–A4 and AX-B1.
OQ-B1: $p = 2$	CLOSED — ZP-B T0	Derived from AX-B1 and MP-1.
OQ-C1: Non-conservatism of DF	CLOSED — ZP-C T2	Infinite sequence divergence proven. No postulates remain.
S1: Distribution stipulation	CLOSED — ZP-C T1	Derived from AX-B1 and RP-1.
OQ-E1: Sequence vs. tree	CLOSED given the occurrence commitment — DA-2	The structure is a forward-directed tree, not a linear sequence. Given that instantiation occurs — the occurrence commitment, not a consequence of T-SNAP — DA-2 supplies the succession and the branching follows. Countable vs. uncountable branching is perspective-dependent (DA-3).
OQ-E2: Cardinality-semilattice correspondence	OPEN	Do specific semilattice structures correspond to specific cardinality regimes? Can the framework make predictions about which instantiations satisfy CH and which do not?

Item	Status	Description
Remaining axioms	INTENTIONAL — AX-B1, AX-G1, AX-G2 (named); A4, quine_unique, bot_self_mem, botCode_is_quine (typeclass fields)	Named commitments: AX-B1 (binary existence), AX-G1 (initial object), AX-G2 (source asymmetry). Structural commitments as typeclass fields: A4/bot_join (ZPSemilattice), AFAStructure.quine_unique and bot_self_mem (the Quine-atom role, which every ZP-A lattice supplies trivially), KleeneStructure.botCode_is_quine (computational closure). Typeclass fields are assumptions carried as hypotheses by the theorems that use them; #print axioms does not surface them. Further commitments are not axioms, among them CC-1 ($S_0 = \perp$), a ZP-A Conditional Claim; T-SNAP's are stated under Premises of T-SNAP (DA-1 insert, § V).
Temperature T in BA-1	PARAMETER — intentional	Universe-contingent. Physical predictions explicitly conditional on instantiation-specific T.

Updated Traceability Register

Claim	Grounded In	Bridge Axiom?	Status
Binary Snap causality	Shape: ZP-C D1, L-RUN, TQ-IH; ZP-A D2; T-SNAP premises: DA-1 insert § V. Occurrence: the occurrence commitment and DA-1	None	Derived for the shape — T-SNAP ✓; the snap occurs given the occurrence commitment and DA-1 (was: Axiomatic — AX-1)
DA-1: Instantiation = execution	DP-2 (TrackedOutput, Snap.lean §VI); ZP-A CC-2, R3; ZP-C L-INF	None	Derived Proposition — primary: DP-2 (da1_minimal_path proved axiom-free in Lean). Informal paths: CC-2/R3 (structural), L-INF (informational), AIT incompressibility (Path 3). AIT+ZF+AFA outside Lean scope.
DA-2: Instantiation succession	ZP-A A4, CC-1; ZP-B C3, R1; T-SNAP	None	Definitional Alignment — clarification of CC-1 scope; no new axiom
C-DA2: Novelty of $\perp$ (conditional)	DA-2, ZP-B C3	None	CONDITIONAL — a modelling commitment. NOT derived: the separateness of the spaces is assumed, not obtained from C3, and snap_arc_z2_loop returns to the same 0.
DA-3: Perspective-relative cardinality	DA-2, C-DA2, ZP-B R1	None	Definitional (DA-3-D1, R-DA3-1); Candidate (DA-3-C1: outside-view inaccessibility)

Claim	Grounded In	Bridge Axiom?	Status
T-SNAP: Snap shape is derived	L-RUN, TQ-IH (the T-BUF chain's lemmas); T-SNAP premises: DA-1 insert § V. DA-1 bears on occurrence, not on the shape	None	Derived — Cross-Framework ✓
AX-1 retirement	T-SNAP proves the shape; the snap occurs given the occurrence commitment and DA-1	N/A	AX-1 is retired: T-SNAP proves the shape, and that the Snap occurs is stated separately: it follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2)
Iterative Forcing T5 (restated)	Selection: succession_succ, ZeroParadox/Ordinal/SnapSuccession.lean § I; for a monotone map sending the tower's stages to c_0 , given the alignment hypothesis $h\varepsilon_0$, nothing below ε_0 fires (snap_unconditional). Iteration: T3, R1, t_snap_irreversible, t_snap_holds_but_nothing_moves	None	On an arbitrary lattice no step is selected. No step is guaranteed.
Multiverse — structural implication	T-SNAP + DA-2 jointly	None	Structural implication given the occurrence commitment — T-SNAP gives the snap's shape, CC-2's undifferentiated $\perp$ the branching, DA-2 the succession
OQ-E2: Cardinality correspondence	DA-3	N/A	OPEN — formal derivation deferred

Validation Status

Component	Status / Notes
DA-1: Derived Proposition (Path 2 recharacterization)	Valid — DP-2 formal core: <code>da1_minimal_path</code> proved axiom-free in Lean (Snap.lean §VI). TrackedOutput separates output value from machine state; pre- and post-instantiation states are provably distinct even when both produce $\perp$. ✓ ZP-K: <code>da1_closed_concrete : IsQuineAtom($\perp$: MachinePhase)</code> proved in Kleene.lean. Path 1 is witnessed by <code>da1_closed_concrete</code> , which proves nothing computational; Path 3's witness is the <code>machinePhaseKleene</code> instance's <code>botCode_is_quine</code> field, a <code>KleeneStructure</code> requirement, not a second independent proof. <code>da1_paths_unified</code> carries the two together as a conjunction of witnesses; that they name one structural fact is the framework's reading. Path 2 (informational bridge, L-INF): FOUNDATIONAL COMMITMENT — a missing principle, not a missing proof. No computability library closes the gap between 'system at P_0 ' and 'system is running.' Forward paths: new axiom, Chalmers' implementation notion, or The Philosophical Question That Started This. ZP-M R-M.1 provides a retrospective structural analysis of why the gap resisted formalization. DA-1 does not depend on Path 2. CC-1 restated in ZP-J as an equivalence, and not forced (a valid sequence can start elsewhere on a carrier with a second point); CC-2 a Forced Metatheoretic Commitment whose Quine-atom role <code>t_exec_iff</code> shows only $\perp$ fills.
T-SNAP: Binary Snap derived	Valid — Derived, in two readings. The Lean form (<code>t_snap_derived</code>) fixes the SHAPE with no hypotheses and no axioms. The seven-step prose argument uses commitments, and that the transition OCCURS is not established by T-SNAP (<code>tsnap_holds_but_nothing_moves</code>). Premises: DA-1 insert § V, Premises of T-SNAP; two of them (CC-1 and occurrence at the first step) as hypotheses, <code>t_snap_given</code> . ✓
AX-1 retirement	Valid — AX-1 is retired, and its content was split in two. T-SNAP proves the shape, strengthened from assumed to derived. That the Snap occurs is stated separately: it follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2). Given CC-1, its first-step form is the hypothesis <code>hocc</code> in <code>t_snap_given</code> ; <code>tsnap_holds_but_nothing_moves</code> shows T-SNAP does not carry it.
DA-2: Instantiation Succession	Valid — Definitional Alignment. Clarification of CC-1 scope. No new axiom. A4 role of $\perp$ extended across instantiation boundaries. ✓
C-DA2: Novelty of $\perp$ (conditional commitment)	CONDITIONAL — a modelling commitment. NOT derived: the separateness of the spaces is assumed, not obtained from C3, which quantifies only over paths WITHIN one space. <code>snap_arc_z2_loop</code> measures against it — the 2-adic arc returns to the same 0. Must not be cited as derived.
Directed instantiation tree	Valid given the occurrence commitment — structural consequence of T-SNAP + DA-2. Once instantiation is taken to occur, branching follows and is not optional. Forward edges only.
Branching tree structure	Valid given the occurrence commitment — T-SNAP + DA-2 jointly. T-SNAP fixes the snap's SHAPE and does NOT establish that it occurs; DA-2 establishes the branching tree structure. Universality (all outbound vectors) is DA-2's scope, not T-SNAP alone.

Component	Status / Notes
Monotonicity and path irrecoverability	Valid — Structural consequence. Monotonicity (T3) constrains direction; additive ontology (R1) prohibits reduction. Path choice is undetermined by algebra.
Ordinal direction of state sequences	Valid — Derived from ZP-A T3 (monotonicity) and ZP-B C3 (irreversibility). Not assumed.
DA-3: Perspective-Relative Cardinality	Valid (definitional components: DA-3-D1, R-DA3-1). Candidate (DA-3-C1: connection to specific set-theoretic independence results). OQ-E2 open.
DA-3-C1 — outside-view inaccessibility	Candidate — no position within any instantiation can replicate the meta-level view of the branching structure. Formal derivation deferred to OQ-E2.
Remaining axioms: AX-B1, AX-G1, AX-G2; typeclass fields A4, quine_unique, bot_self_mem, botCode_is_quine	Named: AX-B1, AX-G1, AX-G2. Typeclass: A4/bot_join (ZPSemilattice), quine_unique + bot_self_mem (AFAStructure), botCode_is_quine (KleeneStructure). Assumptions carried as hypotheses by the theorems that use them; not surfaced by #print axioms. CC-1 is a further commitment (ZP-A Conditional Claim).
ZP-E results T1–T4, T6, T7	Stated in an earlier version of this document (see the repository history) and not restated here. T5 is restated in DA-1 insert § VI.

ZP-E carries three formal inserts (DA-1, DA-2, DA-3). T-SNAP is derived, with its premises stated in the DA-1 insert § V and two of them (CC-1 and occurrence at the first step) as the hypotheses of `t_snap_given`; its irreversibility rests on ZP-A R1 and ZP-B C3, with ZP-G downstream. DA-1 is closed given DP-2 (`da1_minimal_path`, axiom-free). Path 1 is witnessed by `da1_closed_concrete` (ZP-K), which proves `IsQuineAtom( $\perp$  : MachinePhase)` and nothing computational; Path 3's witness is the `machinePhaseKleene` instance's `botCode_is_quine` field, a `KleeneStructure` requirement, not a second independent proof; and Path 2 is a foundational commitment (a missing principle, not a missing proof). Remark R- ϵ_0 justifies the ϵ_0 symbol as a structural analogy. Remark R-AFA rules Foundation out by the self-membership of $\perp = \{\perp\}$ (Regularity, `no_quine_atom`): the forcing and realizability of the AFA requirements are machine-checked (QuineHost), and a narrow Forced Metatheoretic Commitment remains. Open question: OQ-E2. Remaining axioms: AX-B1, AX-G1, AX-G2.